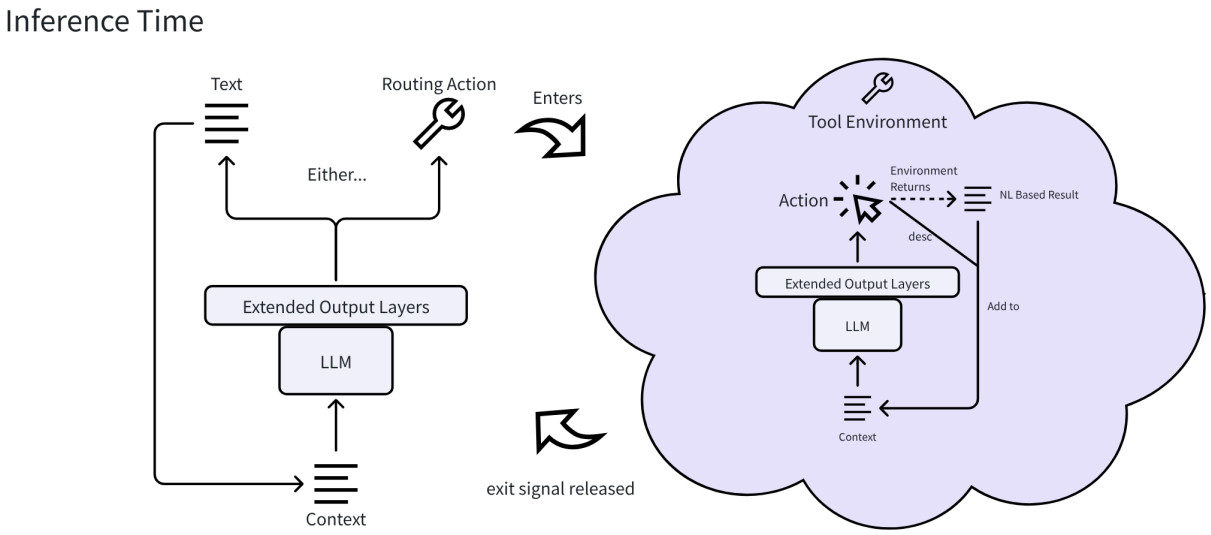
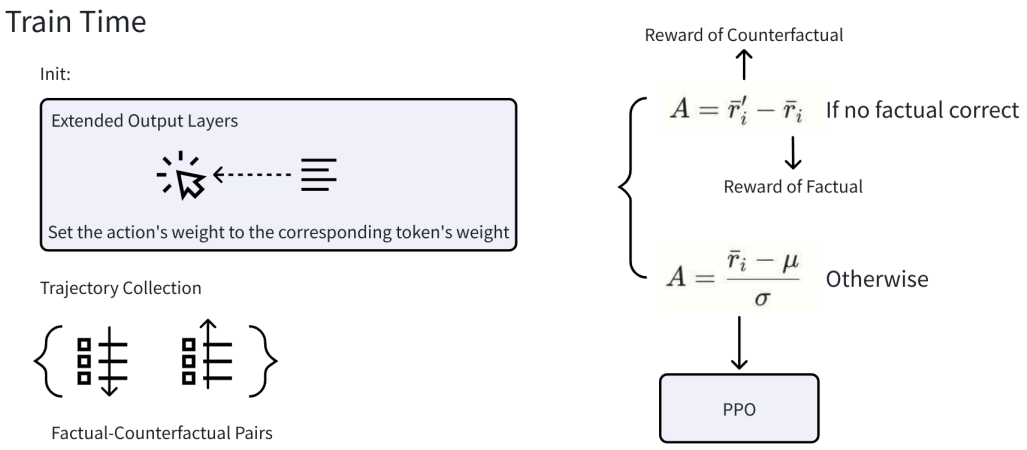
[Class Sharing] EXPANDING THE ACTION SPACE OF LLMS TO REASON BEYOND LANGUAGE

> #Tools #RL

- Graphical Illustration





- What problems does this framework solve?
  - Current tool-use LLMs are overhead: they need to both process the natural language input and need to make tool call decisions, which may degrade their inference quality.
  - Relies on external interpreter: When the model finishes decision, it will output text that related to the tools to be used (for example, MCP's name and parameters). It needs a user-written script to interpret. This is both less robust and cannot be done e2e.
  - The model cannot dynamically change the strategies based on the process in the tool call.
- What does this framework do?
  - Intuition: This framework teaches the model to output actions based on the task in the environment directly.

- The process is defined as a POMDP with states represented as $s_t = (h_t, e_t, z_t)$, where:
  - h_t: context
  - e_t: flag for environment: 0 indicates language space and otherwise tool environment
  - z_t: the latent state of the environment
- Expanded Action space
  - We expand the corpus of the model $\mathcal{V}$ with an action set $\mathcal{E}$ that contains all environment-specific actions of all environments.
  - During the inference time, the model will choose to output either a normal NL token or a routing action g_i to enter a specific tool environment.
  - In the tool environment, the model can **only** output actions available to the environment by using the equation below. The environment will produce feedback, and the feedback will be passed to the model in the NL form. When the environment believes the task is done, it will return an `exit` signal and the pipeline will set e_t to 0.

$$\sigma_e(z)_a = \frac{e^{z_a}}{\sum_{a' \in \tilde{\mathcal{E}}_e} e^{z_{a'}}} 1[a \in \tilde{\mathcal{E}}_e]$$

- ExpA Reinforcement Learning
  - Initialization: At the beginning, the action will gain weight that is the same as the pretrained weight of the corresponding NL token. Formally: $\pi_\theta(a|h_t, e_t) \approx \pi_\theta(\text{desc}(a)|h_t, e_t)$.
  - Counterfactual Policy Optimization
    - Target:

      $$\mathcal{J}_{CPO}(\theta) = \mathbb{E}_{s_0, \mathcal{T}(s_0) = \{(\tau_i, \tau_i')\}_{i=1}^m} \left[\frac{1}{m} \sum_{i=1}^m U_i(\mathcal{T}(s_0); \theta) \right]$$
    - For a normally generated trajectory τ_i, we sample a point with **probability proportional** to the model's likelihood of outputting the tool's description token
    - At this moment, we force the model to enter the tool environment.
    - Then, let the model continue the generation. Such trajectory is called a counterfactual trajectory, denoted as τ_i'.
    - We define U_i:

      $$U_i(\mathcal{T}(s_0); \theta) = \begin{cases} f(\tau_i', \bar{r}_i' - \bar{r}_i; \theta), & \text{if } \bar{r}_j \leq 0 \; \forall j \in \{1, \ldots, m\}, \\ f(\tau_i, \frac{\bar{r}_i - \mu}{\sigma}; \theta), & \text{otherwise.} \end{cases}$$
      - Meaning: If all the normal trajectories fail (reward $\leq$0), we use the **counterfactual trajectory** reward, specifically $\bar{r}_i' - \bar{r}_i$, as the advantage in order to teach the model "see, if you use the tool, it would be better); otherwise, just like GRPO, we calculate the σ and μ of the **factual** trajectories and use the normalized **factual** trajectory as the advantage.
    - The low-level of the RL algorithm is PPO+KL, but the advantage is coming from the mechanism described above instead of the value model+GAE.

- [Paper Review]
- Experiment Results
  - Calc-Bench (including Countdown, etc.): EARL significantly outperforms strong baseline models constrained by pure vocabulary actions (such as SFT+GRPO, Prompt+GRPO), achieving up to a 17.55% improvement in absolute accuracy in the Countdown game. This demonstrates its exceptional strength in scenarios requiring dynamic policy adjustments.

    - Sorting Task: In this highly challenging blind sorting environment (only allowing compare and swap actions), the model not only achieved 100% accuracy on 4-element sorting (Sort-4) but also autonomously discovered an extremely efficient algorithmic structure comparable to classical sorting algorithms.
- Work Cited
    - Paper
    - NotebookLM
    - Chat History